Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/20/2013
PI Name:	Collins, Cynthia Ph.D.		
Project Title:	Gravitational effects on biofilm formation during spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	518-276-4178
Organization Name:	Rensselaer Polytechnic Institute		
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PI Web Page:			
City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	21
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	12/15/2009	**End Date:**	05/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-132 ; STS-135 ; ISS NOTE: Extended to 5/31/2014 per A. Chu/ARC and NSSC information (Ed., 5/22/13) NOTE: Extended to 5/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/1/2012) NOTE: to be extended per A. Chu/ARC (Ed., 1/12/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute) Dordick, Jonathan (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX09AI70G-Flight portion		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	There is an urgent need to understand the effects of microgravity on the growth, cellular physiology, and cell-cell interactions in microbial biofilms. This information can then be used to curtail harmful activities of microbial consortia thriving as biofilms on the International Space Station and for the long-term success of human space exploration. Bacterial biofilms were abundant on the MIR space station and were responsible for increasing corrosion and blocking a water purification system. Health and safety hazards linked to the development of biofilms are also of great concern, including increased rates of infection due to the enhanced survival of organisms within biofilms and increased resilience of biofilm populations to antimicrobial compounds. Furthermore, experiments with Escherichia coli have indicated that biofilms formed under microgravity may have even higher levels of resistance to antimicrobials. We propose to study how gravity alters biofilm formation with the goal of developing new strategies to reduce their impact on the operation of spacecrafts and the health of their crew. We will focus on two model microorganisms that form biofilms both inside and outside of the human body, Pseudomonas aeruginosa and Staphylococcus aureus. These microbes can switch between benign and pathogenic interactions with humans and may be relevant to crew health during extended missions. This project consists of a set of experiments to study biofilm formation during spaceflight that were conducted on Space Shuttle flight STS-135, which launched in July 2011. We examined how the formation and the three-dimensional structures of biofilms formed by these microbes differ in spaceflight compared to normal gravity. We will employ newly developed, non-leaching, nanotechnology-based, antimicrobial surfaces to aid in assessing differences between biofilm formation and structure in normal and microgravity and to assess their potential to reduce the impact of biolfims in future spacecraft designs. NOTE that this is the Flight portion of project entitled, "Gravitational effects on quorum sensing and biofilm formation" and a follow-up set of experiments to those conducted aboard STS-132.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	According to the Center for Disease Control (CDC), hospital-acquired infections are the fourth leading cause of death in the United States behind stroke, cancer, and heart disease. Furthermore, it is estimated that >65% of all bacterial infections are associated with biofilms. A greater understanding of biofilms is essential if we are to find effective methods to combat their formation. Furthermore, the low-shear conditions microbes experience in microgravity are similar to those found in the human body that are difficult to study. This work may provide new insights into the role of shear and other physical effects, such as convection, on biofilm formation.
Task Progress:	We recently published a paper describing the results of our assays of biofilm formation by Pseudomonas aeruginosa during spaceflight in PLoS One. We observed that this organism both formed more biofilm during spaceflight relative to normal gravity controls and exhibited a novel structure. This work garnered significant press attention, including articles on space.com and Popular Science. We have submitted a second paper (under revision) describing how planktonic P. aeruginosa were affected by the microgravity environment. Additional works describing mathematical modeling and S. aureus biofilm formation during spaceflight will be published in the next year.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Collins CH. "Gravitational effects on microbial biofilm formation: implications for spacecraft and crew health." 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, July 16-18, 2013. Oral presentation. 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, July 16-18, 2013. Conference abstract book. , Jun-2013
Abstracts for Journals and Proceedings	Kim W, Tengra FK, Solanki K, Pangule R, Dordick JS, Plawsky JL, Collins CH. "Spaceflight affects gene expression and biofilm formation in Staphylococcus aureus." Poster presentation by W. Kim at the 6th ASM Conference on Biofilms, Miami, FL, September 29-October 4, 2012. 6th ASM Conference on Biofilms, Miami, FL, September 29-October 4, 2012. Conference abstract book, p. 140-141. http://www.asm.org//images/Meetings/2012.9.29.pdf , Sep-2012
Abstracts for Journals and Proceedings	Kim W, Tengra FK, Dordick JS, Plawsky JL, Collins CH. "Biofilm growth by Pseudomonas aeruginosa during spaceflight leads to increased biomass, altered gene expression, and novel structure formation." Poster presentation by C.H. Collins at the 6th ASM Conference on Biofilms, Miami, FL, September 29-October 4, 2012. 6th ASM Conference on Biofilms, Miami, FL, September 29-October 4, 2012. Conference abstract book, p. 209. http://www.asm.org//images/Meetings/2012.9.29.pdf , Sep-2012
Articles in Peer-reviewed Journals	Kim W, Tengra FK, Young Z, Shong J, Marchand N, Chan HK, Pangule RC, Parra M, Dordick JS, Plawsky JL, Collins CH. "Spaceflight promotes biofilm formation by Pseudomonas aeruginosa." PLoS One. 2013 Apr 29;8(4):e62437. http://dx.doi.org/10.1371/journal.pone.0062437 ; Print 2013. PubMed PMID: 23658630 , Apr-2013
Awards	Kim WS. "Peter Gilbert Poster Award for Excellence in Innovation and Biofilm Control for: Kim W. Spaceflight affects gene expression and biofilm formation in Staphylococcus aureus. 6th ASM Conference on Biofilms, Sept 2012." Sep-2012
Dissertations and Theses	Kim WS. "Gravitational effects on Pseudomonas growth and biofilm formation." Dissertation, Rensselaer Polytechnic Institute, April 2013. , Apr-2013
Significant Media Coverage	Howell E. "So You Wanna Go To Space. Can You Put Up With The Superpower Bacteria? Online article on Popular Science's website highlighting results from PI's PLOS ONE paper." PopSci website, posted 6/26/2013: http://www.popsci.com/science/article/2013-06/so-you-wanna-go-space-can-you-put-superpower-bacteria , Jun-2013
Significant Media Coverage	Kramer M. "Bacteria in space grows in strange ways. Online article on SPACE.com website highlighting results from PI's PLOS ONE paper. Note that this article was picked up by many other mainstream news organizations (CBS, Huffington Post, NBC, Yahoo)." SPACE.com Website, posted 7/8/2013: http://www.space.com/21886-space-bacteria-grows-strange-ways.html , Jul-2013